

Regional S-type granites in the Ecuadorian Andes: Possible remnants of the breakup of western Gondwana

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(Received July 1992; Revision Accepted December 1992)

Abstract—Reconnaissance geological mapping of the Ecuadorian Cordillera Real has established the presence of a previously unrecognized regional suite of variably deformed granitoids for which poorly constrained Rb-Sr whole-rock data indicate a minimum Early Jurassic age of 200 ± 12 Ma (initial ratio = 0.7120). This suite, which is associated with low- to medium-grade, semi-pelitic metamorphic rocks, is dominated by peraluminous monzogranites containing biotite $\pm$ garnet $\pm$ muscovite. Geochemically, these granites are S-types and can be readily distinguished from juxtaposed I-type granitoids of the Middle-Upper Jurassic Zamora, Abitagua, and Azafrán batholiths located immediately to the east. Intrusion of these S-type granites may be related to the breakup of western Gondwana

Resumen—El reconocimiento del mapeamiento geológico de la Cordillera Real Ecuatoriana ha establecido la presencia de un conjunto, previamente no identificado, de granitoides variablemente deformados para los cuales las dataciones de roca total de Rb-Sr pobremente registrados indican una edad mínima de Jurásica Inferior de 200 ± 12 Ma ($R_i = 0.7120$). El conjunto, asociado con diversas rocas semi-pelíticas de bajo a mediano grado de metamorfismo, es dominado por monzogranitos peraluminicos con biotita $\pm$ granate $\pm$ muscovita. Geoquímicamente, estos granitos son de “tipo S” y pueden ser fácilmente distinguidos de los granitoides yuxtapuestos “tipo I” de los batolitos Zamora, Abitagua y Azafrán de edad Jurásica Medio-Superior encontrados inmediatamente hacia el este. Se considera la posibilidad de que los granitos “tipo S” puedan ser relacionados a la ruptura de Gondwana occidental.

INTRODUCTION

THE CORDILLERA REAL represents the easternmost of two cordilleras that make up the Ecuadorian sector of the Northern Andes. Throughout its 650-km length, the Cordillera Real is crossed by only five roads. Limited access, high altitude (the main watershed lies between 5800 and 3200 m), and heavy rainfall combine to discourage fieldwork, so that the geology of the area has remained poorly known until recently.

In 1986, a bilateral Ecuadorian-British Technical Cooperation Program was initiated to carry out a regional investigation into the nature and economic mineral potential of the metamorphic rocks that comprise the bulk of the Cordillera Real. Based on about 10 man-years of field-oriented studies, supported by geochronological (Aspden *et al.*, 1992) and geochemical studies (Litherland *et al.*, 1990), the pre-Cretaceous rocks of the Cordillera have been divided into a series of informal, regional, lithotectonic divisions (Aspden and Litherland, 1992).

Lithologically, the Loja division consists principally of a variably metamorphosed pelitic-psammitic sequence (the Agoyán and Chiquinda subdivisions) and metagranitoids (Fig. 1). In the west are the foliated biotite + garnet $\pm$ muscovite granites of the Tres Lagunas subdivision; in the southeast is the elongate Sabanilla subdivision, a more

heterogeneous unit that is dominated by biotite $\pm$ garnet $\pm$ muscovite-bearing orthogneisses and migmatites.

Although isolated occurrences of these rocks had been noted previously (Kennerley *et al.*, 1973; Harrington, 1957; Colony and Sinclair, 1932), their true regional extent and significance had not been appreciated (*e.g.*, Baldock, 1982). This preliminary contribution gives a brief description of the field and petrographic characteristics of these rocks and presents new geochronological and whole-rock geochemical data. Together, these data indicate that the Loja division granitoids show many features of S-type granites (Chappell and White, 1974) and are quite distinct from “normal” Andean (cordilleran) I-type granitoids (Cobbing, 1990; Pitcher, 1983), as exemplified by the Zamora, Abitagua, and Azafrán batholiths in Ecuador (Fig. 1).

GEOLOGY AND PETROGRAPHY OF LOJA DIVISION GRANITOIDS

Tres Lagunas Granites

Outside of shear zones, the Tres Lagunas granites are petrographically distinctive, normally consisting of medium- to coarse-grained granites with prominent,

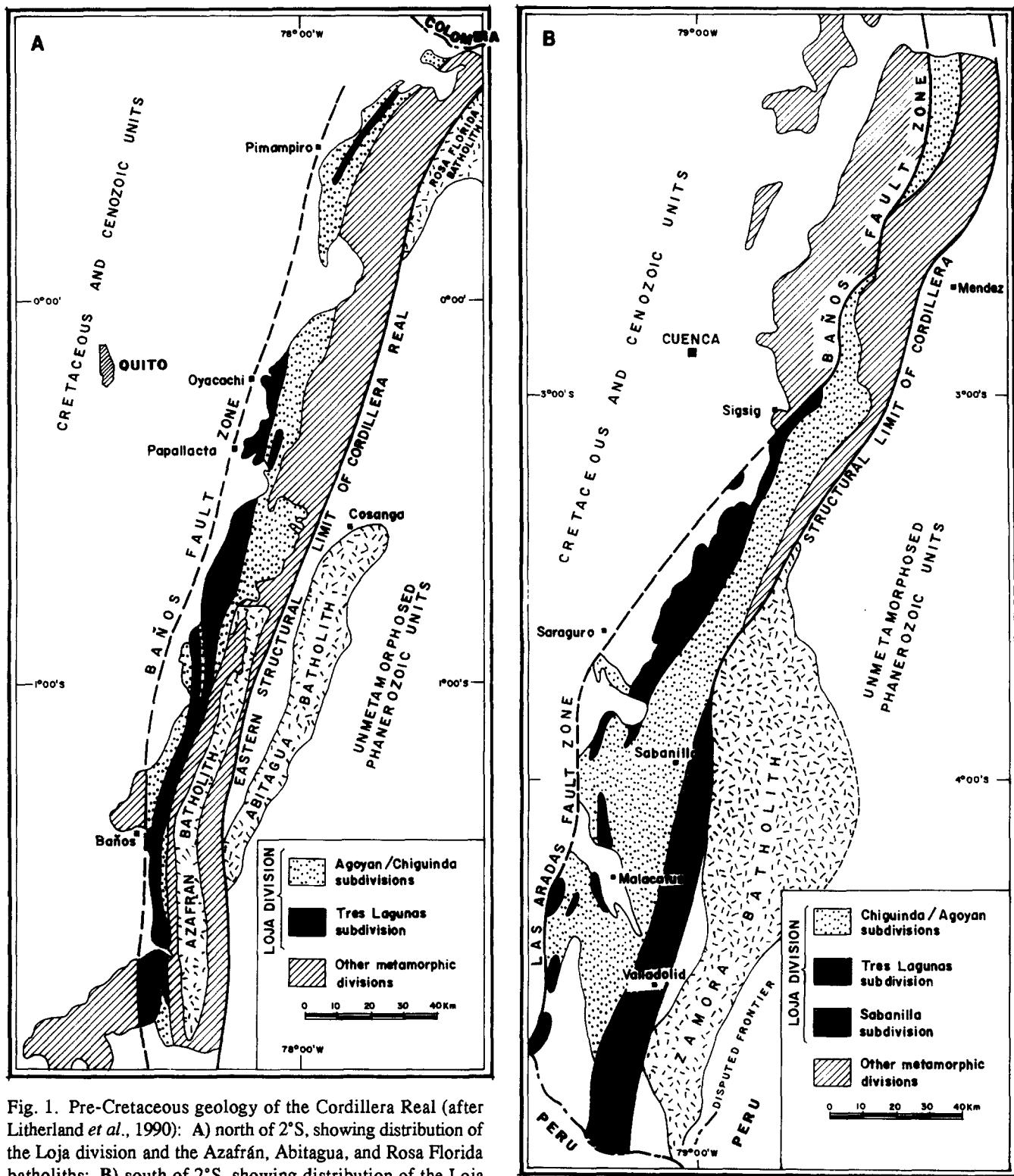


Fig. 1. Pre-Cretaceous geology of the Cordillera Real (after Litherland *et al.*, 1990): A) north of 2°S, showing distribution of the Loja division and the Azafrán, Abitagua, and Rosa Florida batholiths; B) south of 2°S, showing distribution of the Loja division and the Zamora batholith.

smoky-blue to grey alkali feldspar megacrysts, up to 14 cm in length. Many samples contain pale blue quartz crystals of uncertain origin. The major mafic mineral, biotite, is typically reddish-brown in thin section and up to 1 cm in diameter; it may constitute up to 10% of the mode. Hornblende has not been found in these rocks.

The alkali feldspar is normally perthitic, and the plagioclase ranges from albite to oligoclase. Narrow rims of

blue-grey alkali feldspar are relatively common on earlier-formed, often euhedral plagioclase, but “rapakivi” overgrowths also occur. The alkali feldspar megacrysts contain inclusions of cream-colored plagioclase and/or biotite ± quartz.

Garnet is a common accessory mineral, attaining 30% of the mode in a belt of garnet granites between Papallacata and Oyachachi (Fig. 1A). Cordierite has been

recorded but is rare. Muscovite is fairly common but is mostly subordinate to biotite, which it generally replaces. Secondary granoblastic quartz fabrics are widespread and are accompanied by partial replacement of feldspar and mica by quartz. Other late features include the formation of epidote, the sericite-zoisite alteration of feldspars, the recrystallization and/or chloritization of biotite, and the growth of brown tourmaline. Minor amounts of opaque minerals are also present, but their compositions have not been determined directly. However, outcrops show consistently low magnetic susceptibility readings, suggesting the absence of magnetite.

Relatively underformed granites can often be traced into gneissic belts related to vertical or steep, westward-dipping, Andean-trending shear zones in which S-C mylonites (Berthé *et al.*, 1979; Lister and Snoke, 1984) are widely developed. In places, the mylonites are cut by younger, underformed pegmatite veins of quartz $\pm$ tourmaline $\pm$ feldspar $\pm$ muscovite.

Xenoliths are relatively uncommon but include both meta-sedimentary and meta-igneous material. Large xenocrysts of white vein quartz (up to 5 cm) are common in the Malacatus area, and synplutonic amphibolites are present east of Baños (Fig. 1).

Contacts of the Tres Lagunas subdivision are tectonic. North of latitude 2°S, the country rocks (Agoyán subdivision, Fig. 1A) are typically medium-grade, aluminous schists and paragneisses, with rare incipient migmatization, whereas in the south they normally comprise low-grade, semi-pelitic phyllites and quartzites (Chiguinda subdivision, Fig. 1B).

Sabanilla Orthogneisses

Fewer petrographic details are available for the Sabanilla subdivision orthogneisses, which consist essentially of medium-grained, foliated biotite $\pm$ muscovite $\pm$ garnet granites. In contrast to Tres Lagunas, these granites do not contain blue quartz, nor are they commonly megacrystic. They are also more homogeneously foliated, so they may have been deformed at somewhat higher temperatures and possibly deeper levels (Gapais, 1989) — an inference supported by the more common occurrence of migmatites and the occasional presence of kyanite- and sillimanite-bearing assemblages in associated paragneisses. Even away from the more obviously migmatitic parts, the Sabanilla orthogneisses are texturally heterogeneous and often contain meta-sedimentary xenoliths in various stages of digestion. Biotite schlieren and clots are extremely common. In some outcrops, randomly oriented blocks of orthogneiss, apparently similar in composition to the host, have been noted. As with the Tres Lagunas subdivision, relatively small mafic bodies, of amphibolite, are present in some areas (*e.g.*, north and east of Valladolid, Fig. 1B).

Age of Loja Division Granites

The age of the Loja division granites is not precisely known, but blue quartz clasts, presumed to be derived from the Tres Lagunas granites, occur in fossiliferous Lower Jurassic meta-sedimentary rocks exposed south of Baños (Howarth and Ivimey-Cook, 1991) (Fig. 1A). Attempts to date both the Tres Lagunas and Sabanilla subdivisions, using K-Ar (biotite, muscovite), Sm-Nd (garnet/whole-rock), and Rb-Sr (whole-rock) methods, have been generally unsuccessful. At present, the most reliable data come from the Tres Lagunas subdivision, where the combined Rb-Sr whole-rock data (17 points) give an isochron age of 200 ± 12 Ma (MSWD = 169, $R_i = 0.7120$; Fig. 2). Despite the high degree of scatter, this is considered to be the minimum age for emplacement (Aspden *et al.*, 1992).

ANALYTICAL DATA

The material used in this study was collected using a hand-held rock drill and dynamite. In the case of the Tres Lagunas granites, only the more massive (*i.e.*, least foliated) outcrops were sampled. In all, 24 whole-rock analyses representing each of the Loja division granites (LDG) and the Zamora, Abitagua, and Azafrán batholiths (ZAAG) are currently available (Aspden *et al.*, 1990). The Tres Lagunas samples were collected from areas south of Sigsig, east of Saraguro, and north of Malacatus; the Sabanilla subdivision was sampled north of Valladolid and east of Sabanilla (Fig. 1B). Representative analyses from the LDG and ZAAG suites are listed in Table 1.

The Zamora, Abitagua, and Azafrán batholiths, analytical data for which are included here for comparative purposes, form the southern part of a belt of predominantly Middle to Upper Jurassic batholiths that can be traced throughout the Northern Andes (Aspden *et al.*, 1991, 1990, 1987; McCourt *et al.*, 1984). They are typical of Andean I-type granitoids, having a wide range in SiO₂ and high Na₂O values, and are commonly hornblende bearing.

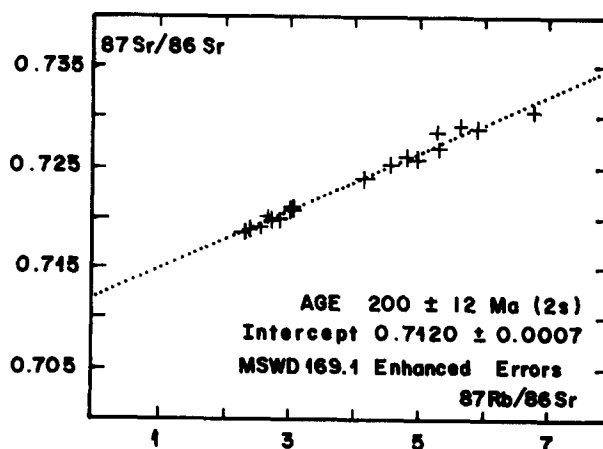


Fig. 2. Rb-Sr whole-rock isochron diagram for the Tres Lagunas subdivision (Aspden *et al.*, 1992).

Table 1. Representative whole-rock analyses of the Zamora, Abitagua, and Azafrán batholiths (ZAAG) and of the Loja division granites, Tres Lagunas and Sabanilla subdivisions (LDG).

Sample:	Z A A G S U I T E						L D G S U I T E											
	CR5A	CR5D	CR6B	CR8A	CR16H	CR21B	CR23A	CR23H	SH9B	SH1A	SH1D	SH1F	SH14A	SH14D	SH14G	SH16A	SH16B	
SiO ₂	77.65	73.28	65.40	76.04	54.70	65.09	71.32	73.21	74.20	72.47	70.20	69.85	68.51	71.88	72.74	73.16	71.01	70.99
TiO ₂	0.06	0.23	0.54	0.12	0.68	0.44	0.74	0.50	0.51	0.51	0.71	0.72	0.75	0.55	0.53	0.45	0.55	0.55
Al ₂ O ₃	12.65	14.12	16.41	13.09	17.41	16.05	12.93	13.22	12.93	14.17	13.67	13.53	14.10	13.55	13.30	13.44	13.67	13.73
Fe ₂ O ₃	0.47	0.88	1.39	0.67	3.08	2.03	0.53	0.57	0.89	0.61	0.95	0.86	0.99	0.76	0.45	0.50	0.65	0.82
FeO	0.29	0.93	2.32	0.47	4.63	2.24	3.70	2.94	2.95	3.23	3.52	3.73	3.80	2.57	2.62	2.20	2.58	2.52
MnO	0.08	0.05	0.09	0.09	0.13	0.08	0.06	0.10	0.08	0.06	0.10	0.10	0.10	0.08	0.07	0.06	0.11	0.09
MgO	0.00	0.33	1.17	0.06	4.90	1.57	1.93	1.36	1.34	1.15	1.65	1.73	1.83	1.03	0.96	0.79	1.15	1.13
CaO	0.38	1.62	3.23	0.83	8.09	4.24	2.23	1.47	0.59	1.04	1.96	1.95	2.26	1.65	1.47	1.41	1.82	1.85
Na ₂ O	3.87	3.82	3.51	3.93	3.37	3.88	2.24	2.30	1.75	2.02	2.46	2.33	2.42	2.53	2.49	2.55	2.45	2.49
K ₂ O	4.47	3.69	5.08	4.36	0.96	2.44	3.03	3.14	3.22	3.28	3.18	3.14	3.28	4.05	4.44	4.53	4.21	3.82
P ₂ O ₅	0.03	0.09	0.19	0.04	0.12	0.14	0.17	0.16	0.14	0.18	0.18	0.17	0.17	0.18	0.19	0.19	0.16	0.17
LOI	0.32	0.65	0.69	0.42	2.02	1.47	1.05	1.03	1.44	1.47	1.54	0.59	1.60	1.23	0.96	1.01	1.32	1.58
Total	100.28	99.69	100.03	100.12	100.98	99.68	99.92	100.00	100.05	100.19	100.11	99.70	99.81	100.06	100.21	100.30	99.68	99.84
Fe ₂ O ₃ T	0.80	1.91	3.97	1.19	8.22	4.52	4.64	3.83	4.17	4.20	4.86	5.01	5.21	3.62	3.36	2.95	3.52	3.62
Cr	24	26	22	17	36	17	58	64	84	77	107	101	99	35	37	32	39	35
Cu	0	4	22	0	35	5	11	19	13	13	23	24	23	14	12	10	13	13
Ga	13	13	17	13	17	16	15	15	15	18	20	17	17	16	14	15	14	15
Nb	10	5	11	12	5	6	14	11	10	12	11	11	13	13	12	10	11	12
Ni	2	3	5	2	27	6	24	19	197	150	212	199	210	13	11	12	13	12
Ph	18	10	20	13	5	6	20	19	25	31	22	22	18	21	27	26	45	33
Rb	163	87	134	103	24	70	125	125	141	139	128	121	134	196	189	192	150	158
Sr	24	261	432	90	270	368	206	127	76	118	146	147	171	95	104	98	132	114
Th	17	4	5	9	0	8	9	7	7	10	14	13	11	15	12	10	12	13
V	7	22	52	10	211	91	103	78	84	87	100	94	107	57	57	52	65	55
Y	18	11	22	13	24	16	29	23	21	26	33	34	31	41	40	34	32	35
Zn	23	41	53	32	46	32	73	65	66	67	78	77	81	63	52	52	51	60
Zr	71	111	207	95	71	93	258	183	135	157	193	200	210	185	177	155	178	175
Ba	144	1515	1218	747	307	842	589	651	1007	582	818	861	844	535	659	756	1024	660
La	2	10	27	5	3	6	20	9	9	12	29	29	20	12	18	15	21	15
Ce	29	10	38	8	16	6	51	33	29	42	54	58	66	64	56	38	35	73
Nd	8	5	22	5	17	11	19	22	25	23	23	19	35	30	30	18	12	47
Cl	227	204	373	202	560	420	129	93	207	114	104	92	104	145	113	166	133	163
S	80	86	150	82	217	99	133	353	93	369	884	1275	481	351	316	223	298	306

Analyses were carried out by x-ray fluorescence spectrometry at the University of Keele, UK. Key to samples (see Fig. 1 for general locations, precise locations available upon request): CR5A, CR5D, CR6B — Abitagua batholith; CR8A — Azafrán batholith; CR16H, CR21B — Zamora batholith; CR23A, CR23H — Sabanilla subdivision east of Sabanilla; SH9B, SH1F — Sabanilla subdivision north of Valladolid; SH11A, SH11D, SH11F — Tres Lagunas subdivision south of Sigsig. Accuracy quoted for major-element values is 1–1.5%, except for SiO₂ and Fe₂O₃, which are 0.5%. For trace-element values > 100 ppm the accuracy is <5%, but for values > 10 ppm the accuracy increases to ~10%. Fe₂O₃ and FeO were analyzed using dichromate titration.

Initial $^{87}\text{Sr}/^{86}\text{Sr}$ ratios for the Abitagua and Zamora batholiths range from 0.7037 to 0.7056 (Aspden *et al.*, 1992) and are similar to values of 0.7034–0.7048 obtained by Brook (1984) from the Ibagué batholith in Colombia. These data suggest derivation from a fairly primitive isotopic source. Consequently, the entire belt is interpreted to represent the principal magmatic products of Jurassic subduction along the paleocontinental margin of northwestern South America (Aspden *et al.*, 1987).

Geochemistry

Based on calculated CIPW normative values, the majority of the LDG plot in the quartz-rich (normative quartz > 38%) part of the monzogranite field in the QAP ternary diagram. In contrast, the ZAAG show a greater compositional range, including not only monzogranites but also granodiorites and more basic dioritic variants (Fig. 3).

Furthermore, variation diagrams reveal clear differences between the LDG and ZAAG suites. When plotted against SiO_2 , the LDG suite stands out as enriched in various elements, including TiO_2 , P_2O_5 , Cr, and Zn (Fig. 4). Some elements that have concentrations close to their ZAAG counterparts of comparable SiO_2 content show a quite different variation trend. Thus, whereas Th, Ce, Y, and Nb generally rise with increasing SiO_2 in the I-type ZAAG suite, they appear to fall with increasing SiO_2 in the S-type LDG suite (Fig. 5). Based on this, we conclude that the LDG are indeed a separate group and could not have formed by fractionation of ZAAG-type magmas. This contrasts with the interpretation of the more evolved parts of the Cordillera Blanca batholith in Peru, which exhibit many S-type features, as having been derived from an I-type parent (Atherton and Sanderson, 1987).

Differences between the LDG and ZAAG suites can also be seen on a number of other plots. In both the K_2O vs Na_2O and the $\text{Al}/(\text{Na}+\text{K}+\text{Ca}/2)$ vs SiO_2 diagrams (Figs. 6 and 7), the peraluminous ($\text{A}/\text{NKC} > 1.1$) LDG suite and the metaluminous ($\text{A}/\text{NKC} < 1.1$) ZAAG suite fall within the S- and I-type fields, respectively, with little or no overlap between the groups. Equally, on the ACF plot (Fig. 8) the S-type LDG straddle the plagioclase-biotite tie line and extend into the Al-rich part of the diagram, whereas the I-type ZAAG straddle the plagioclase-hornblende tie line, with some points lying on the CaO-rich side.

The Chappell and White (1974) classification of granites into S- and I-types is broadly similar to the ilmenite- and magnetite-series of Ishihara (1977) in that all S-types belong to the ilmenite-series and the majority (but not all) of the I-types correspond to the magnetite-series (Beckinsale, 1979). Magnetite- and ilmenite-series granites can be distinguished using an $\text{Fe}_2\text{O}_3/\text{FeO}$ vs SiO_2 plot (Lehmann and Harmanto, 1990) — a diagram that also clearly separates the LDG and the ZAAG suites and classifies them as belonging to the ilmenite- and magnetite-series, respectively (Fig. 9).

The above data illustrate that the LDG and ZAAG suites represent two distinct groups of granites and that the

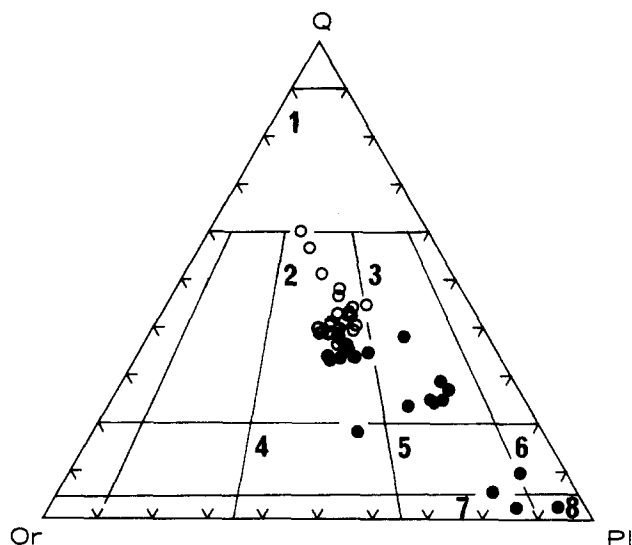


Fig. 3. QAP ternary diagram (after Streckeisen, 1976) based on CIPW normative values (Q = quartz, Or = orthoclase, Pl = anorthite + albite). Open circles, Loja division granites (LDG; Sabanilla, Malacatus, Peggy, Saraguro, and Valladolid areas); closed circles, ZAAG granitoids (Zamora, Abitagua, and Azafrán batholiths). Key to numbers: 1, quartz-rich granitoids; 2, monzogranite; 3, granodiorite; 4, quartz-monzonite; 5, quartz-monzogranite/gabbro; 6, quartz-diorite/gabbro; 7, monzodiorite/gabbro; 8, diorite/gabbro.

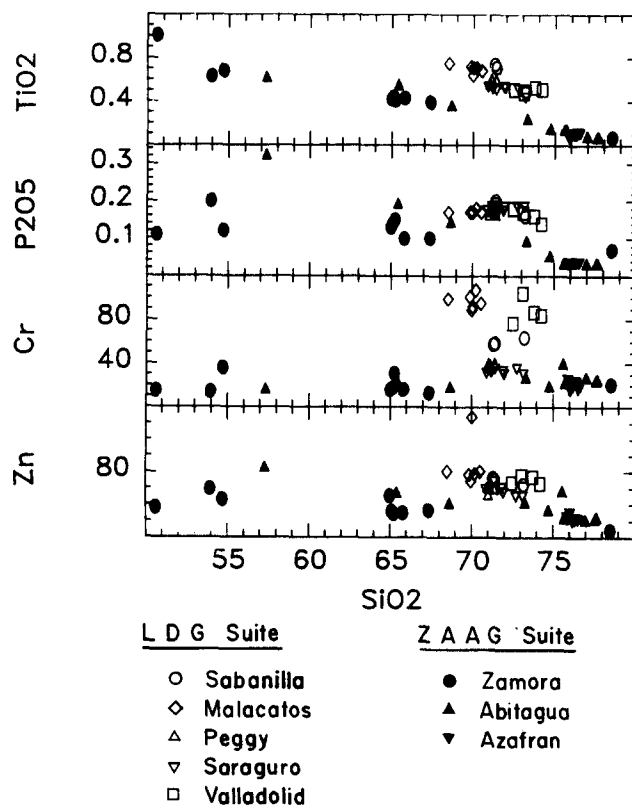


Fig. 4. Harker diagrams showing TiO_2 , P_2O_5 , Cr, and Zn vs SiO_2 for LDG and ZAAG granitoids.

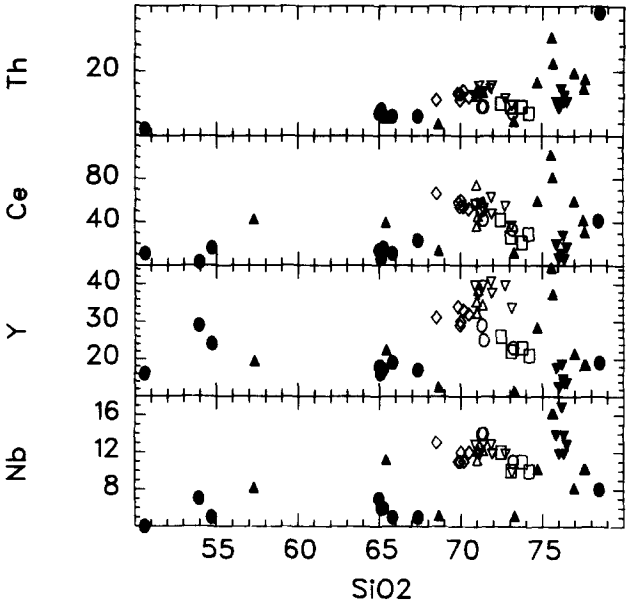


Fig. 5. Harker diagrams showing Th, Ce, Y, and Nb vs SiO_2 for LDG and ZAAG granitoids.

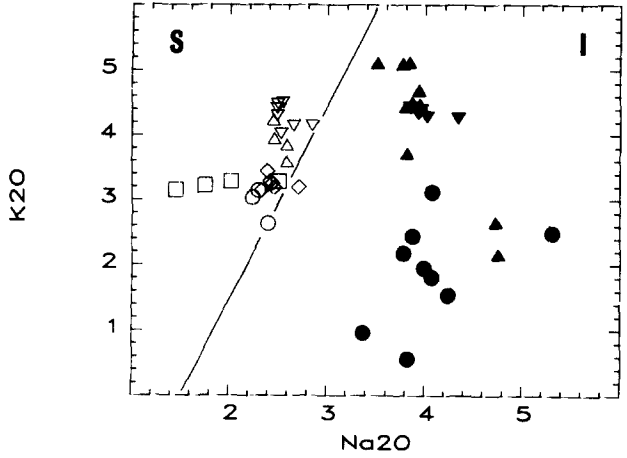


Fig. 6. K_2O vs Na_2O diagram for the LDG and ZAAG granitoids. I-type and S-type granite fields after Chappell and White (1974). Symbols as in Fig. 5.

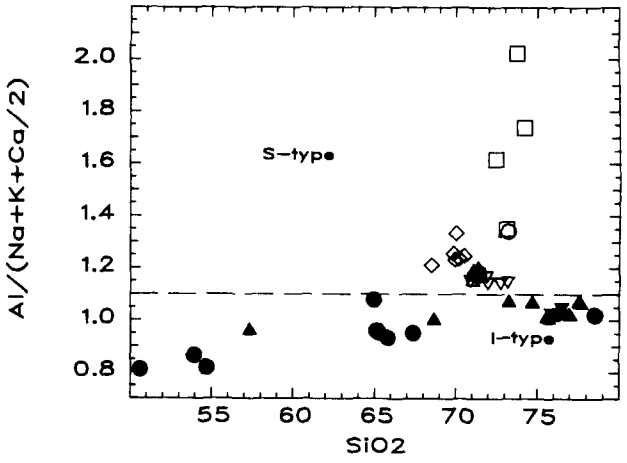


Fig. 7. Aluminosity index vs SiO_2 for the LDG and ZAAG granitoids. I-type and S-type granite fields after Chappell and White (1974). Symbols as in Fig. 5.

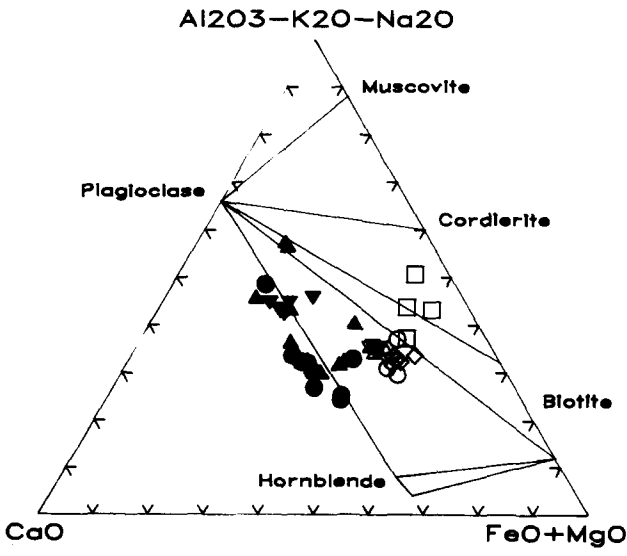


Fig. 8. ACF ternary diagram for the LDG and ZAAG granitoids. Symbols as in Fig. 5.

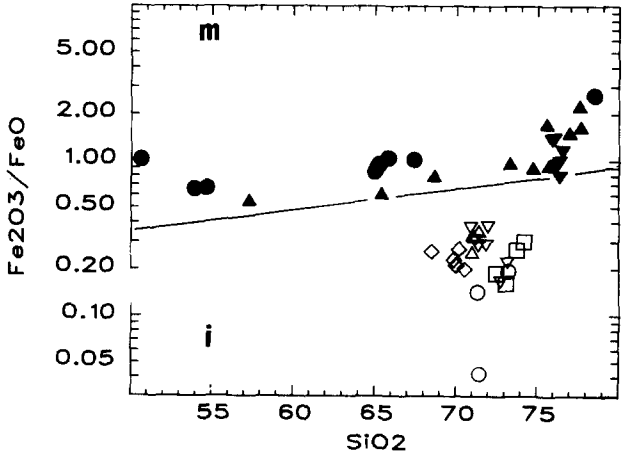


Fig. 9. $\text{Fe}_2\text{O}_3/\text{FeO}$ vs SiO_2 plot for the LDG and ZAAG granitoids. Fields of magnetite- (m) and ilmenite- (i) series granites after Ishihara *et al.* (1979) and Lehman and Harmanto (1990). Symbols as in Fig. 5.

LDG have many of the geochemical characteristics of S-type granites (see Pitcher, 1987, 1983; Hine *et al.*, 1978; Chappell and White, 1974). LDG samples collected from different areas appear to form closely clustered compositional subgroups on a number of plots. These subgroups are apparent in diagrams having Cr, Ni, and SiO_2 as discriminants (Figs. 4 and 10) and, although data are limited, they suggest a lack of regional uniformity within the LDG suite. Each subgroup appears to be geochemically distinct, and we further suggest that the consistency of "mobile" elements, such as Rb (Fig. 11), argues that, overall, the compositions of the LDG have not undergone major chemical modification either by late-stage alteration or during regional shearing (mylonitization).

Although detailed petrogenetic modeling is beyond the scope of this paper, White and Chappell (1977) suggested that variation trends in granites of crustal origin may arise by mixing of different melt and restite proportions. In the case of the LDG suite, this model may, for example, account for the trends of decreasing Y, Nb, Ce, and Th observed in Fig. 5. These elements could be envisaged as remaining concentrated in restite phases such as ilmenite, garnet, and monazite and falling to very low concentrations in the separated eutectic melt (with *ca.* $\text{SiO}_2 = 76\%$). However, other elements, including Ti, P, Cr, and Zn, differ in that their decreasing trends can be extrapolated to reach zero at $\text{SiO}_2 > 80\%$ (Fig. 4), with significant concentrations remaining at 76% SiO_2 . This suggests that part of these elements may have entered the melt, presumably by melting of mafic silicates and apatite, while a proportion was retained in the more refractory phases.

DISCUSSION AND CONCLUSIONS

Geochemically, the LDG suite can be classified as S-type granites, and their relatively high $^{87}\text{Sr}/^{86}\text{Sr}$ ratios (Fig. 2) also suggest involvement of a substantial crustal component in their origin. This value (0.7120) is considerably greater than the entire range of Ri values obtained for the ZAAG granitoids and is similar to that of crustally contaminated modern andesties in Colombia (James, 1984).

Although regional S-type granites have not previously been reported from the Northern Andes, "S-like" granites, principally of Permo-Triassic age, are present in both the Central and Southern Andes, where they are generally considered to have been emplaced in extensional settings, possibly related to crustal relaxation preceding the breakup of Gondwana (Avila-Salinas, 1990; Suárez *et al.*, 1990; Rapela *et al.*, 1989; Kontak *et al.*, 1985). The possibility therefore exists that the LDG could represent equivalents of the "S-like" granites in Peru, Bolivia, and northern Chile. In marked contrast to these occurrences, however, the LDG suite is regionally developed and typically has a strong mylonitic fabric. Equally, these granites do not appear to be genetically related to I-types (*i.e.*, the ZAAG suite) nor, as far as we can tell, do they occur in batholiths of mixed S-I character (*cf.* Avila-Salinas, 1990;

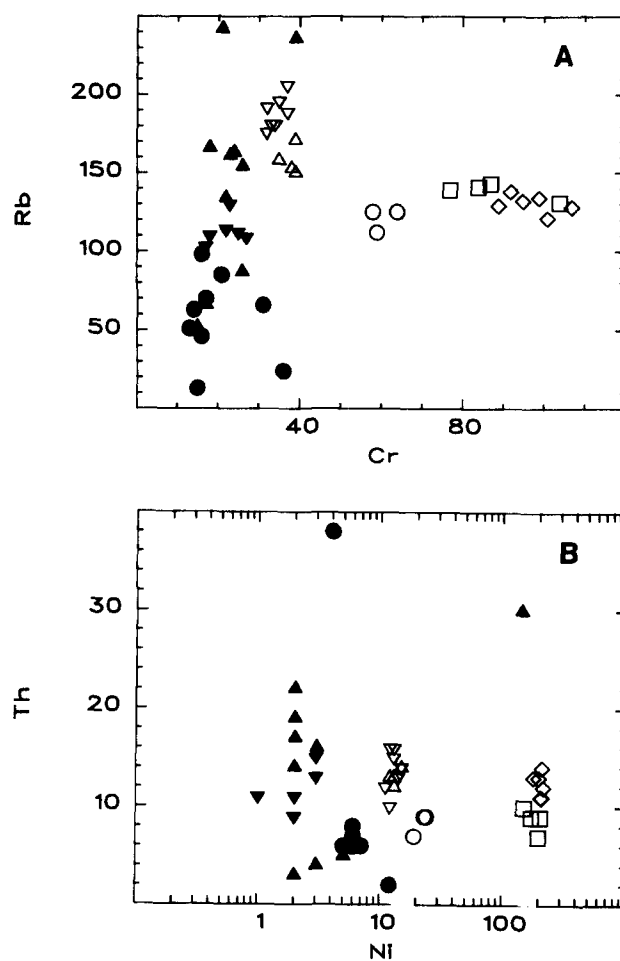


Fig. 10. Rb vs Cr (A) and Th vs Ni (B) diagrams for the LDG and ZAAG granitoids. Symbols as in Fig. 5.

Cobbing, 1990; Rapela *et al.*, 1989; Atherton and Sander-son, 1987).

Pitcher (1987, 1983) suggested that different types of granites could be related to different tectonic environments. Specifically, he considered the development of regional S-type granites as characteristic of zones of continental collision and also encratonic ductile shear belts. More recently, Cobbing (1990), although agreeing that S-type granites with crustal signatures are associated with collisional settings, has emphasized the diversity of crustal granites and pointed out that similar granites can be developed in a variety of tectonic settings. In his view, the composition of the crustal source region is of prime importance in determining granite type and, ultimately, the relative amounts of mantle-derived and crustal materials that are mobilized during magma genesis.

Pearce *et al.* (1984; see also Brown *et al.*, 1984) proposed that the tectonic setting of most granites could be determined according to the abundance of certain trace elements. Using a Rb vs Nb+Y plot, Pearce *et al.* (1984) distinguished between granites generated in volcanic-arc (VAG), syncollisional (Syn-COLG), and within-plate (WPG) settings. On their plot, all the Ecuadorian samples would be classified as volcanic-arc granites. However, it

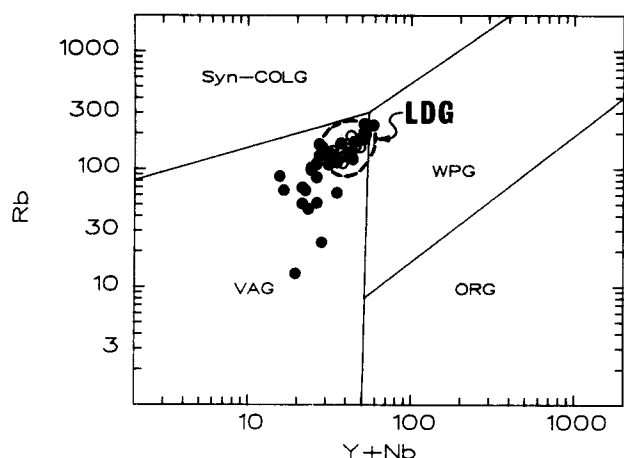


Fig. 11. Rb vs Y+Nb discriminant plot (Pearce *et al.*, 1984): VAG, volcanic-arc granites; WPG, within-plate granites; Syn-COLG, syn-collisional granites; ORG, ocean-ridge granites; open circles, LDG; closed circles, ZAAG.

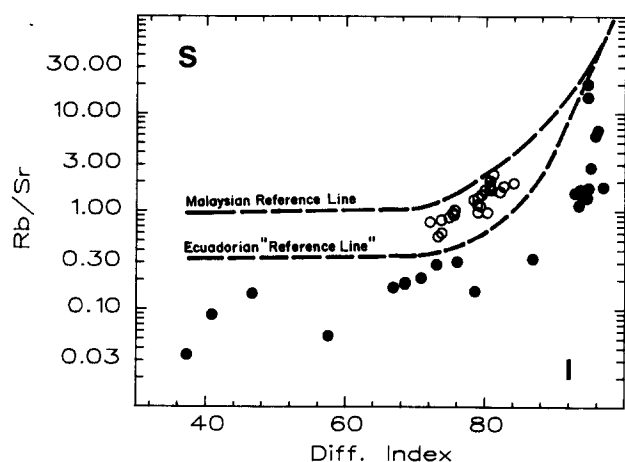
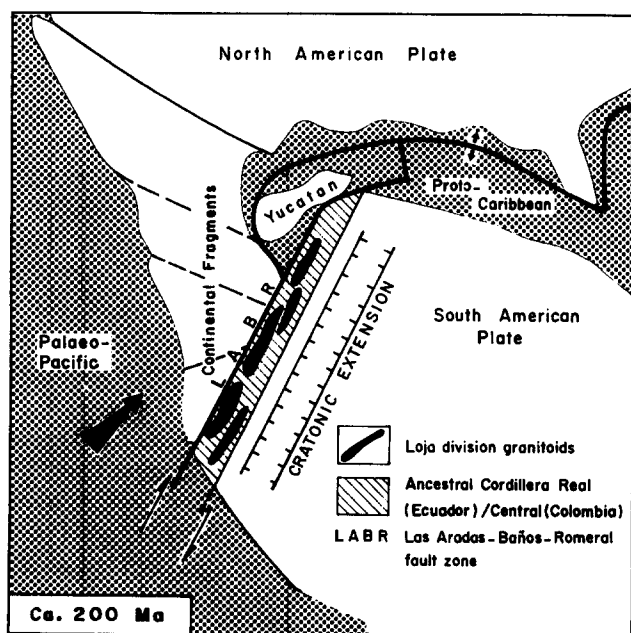


Fig. 12. Rb/Sr vs DI (Thorton and Tuttle differentiation index) discriminant plot, I- and S-type granite fields after Pitfield (1988) and Cobbing (1990). Open circles, LDG; closed circles, ZAAG.



may be significant that the Ecuadorian S-types plot in the upper part of the VAG field, close to where these granites converge with WPG and Syn-COLG (Fig. 11). Compared with other S-types, the LDG suite is relatively poor in Rb and, although not proven, we assume that this is more likely to reflect the composition of the source area rather than being diagnostic of a particular tectonic setting (see also Cobbing, 1990; Chappell and Stephens, 1988). Indeed, the more "primitive" nature of the LDG suite with respect to Southeast Asian S-type granites can be seen in Fig. 12. On this diagram, most Ecuadorian samples plot in the I-type field (*i.e.*, below the Malaysian reference line) but, as Cobbing (1990) pointed out (see also Pitfield, 1988), the Malaysian reference line is empirical and may not serve to distinguish I- and S-type granites from elsewhere in the world. Nevertheless, although the LDG do not achieve Rb/Sr ratios > 1.0 at DI values $< ca. 85$ (as is the case in Southeast Asian S-types), they do have consistently higher Rb/Sr ratios at the same DI than those of the ZAAG.

In a review of the early Mesozoic history of the Northern and Central Andes, Jaillard *et al.* (1990) suggested that the Late Triassic-Liassic separation of North and South America, between the paleo-Mexican margin and what is now part of the Northern Andes, included rift-related extension and possible transcurrent (?transpressional) stress. This model accounts for the early Mesozoic "extensional" regime preserved in the geological record of Colombia and Ecuador and, in our view, may also explain the presence of a regional belt of variably deformed S-type granites and migmatites. We assume that the highly oblique approach of the paleo-Pacific plate resulted in a significant amount of transpressional strike-slip (Fig. 13) and possibly limited subduction along the northwestern margin of South America. D'Lemos *et al.* (1992; see also Hutton, 1988; Wickham and Oxburgh, 1986; Pitcher, 1983) have recently proposed that crustal-scale transpressional shear zones provide settings favorable for generating and emplacing S-type (anatectic) granites. In Ecuador, S-C mylonites (Lister and Snoke, 1984) are widely developed throughout the Cordillera Real, and it has been suggested that repeated episodes of dextral transpression affected the Cordillera during Mesozoic time (Aspdén and Litherland, 1992).

The western (tectonic) limit of the Loja division coincides with the Las Aradas-Baños fault (Fig. 1), a line of intense shearing and regional mylonite development, which we suggest represents the remnants of the zone of separation between the paleo-Mexican margin and northwestern South America. Farther to the north in Colombia, we would correlate the Las Aradas-Baños fault with the Romeral fault zone (Fig. 13). If such a correlation is valid, then one would predict that, as geological exploration of

Fig. 13. Sketch of the proposed Late Triassic-Liassic (dextral) transpressional regime affecting the margin of northwestern South America (modified from Jaillard *et al.*, 1990), illustrating the possible tectonic setting of Loja division granitoids.

the Colombian Central Cordillera continues, granites equivalent to those of the Loja division will be recognized. At present, the most probable correlatives are a small group of plutons described mainly from Antioquia and Caldas in the central part of the Cordillera, immediately to the east of the Romeral fault zone (e.g., Puqui, El Buey, and Amago stocks). These plutons typically have faulted contacts and a strong, tectonic, biotite fabric; although poorly dated, they are considered to be Triassic in age (Jaillard *et al.*, 1990; Aspden *et al.*, 1987; Macía and Mojica, 1981). According to Hall *et al.* (1982), the Puqui stock, which has K-Ar muscovite and biotite ages ranging from 239 ± 7 to $211 \pm ?$ Ma (see also Botero, 1975), has "gradational" and, in part, migmatitic contacts with the surrounding micaceous, gneissose, host rock and is interpreted to be syntectonic and of possible crustal origin.

Acknowledgments—This paper is published with permission of the Director of the British Geological Survey (NERC) and the Instituto Ecuatoriano de Minería (INEMIN). Work in Ecuador and in the UK was carried out as part of an ongoing bilateral technical cooperation project between the governments of Ecuador and the UK (via the Overseas Development Administration). Special thanks are due Srs. Casanova and Céleri of INEMIN. We are grateful to R. J. Cobbing and R. J. Pankhurst for their comments on an early draft of this paper, and to Profs. Duque and Equez for helpful suggestions.

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